

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

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Location: MSB 104



Lecture 08
Clustering
(k-mean, Hierarchical, DBSCAN)

Today's class:

- *What is clustering?*
- *k-means clustering*
- *Hierarchical clustering*
- *DBSCAN*

Where We Left Off

Every Method So Far Had an Answer Key

- Decision trees, naive Bayes, k-NN, logistic regression, SVM.
- Five different ideas. All five answered the same question.
- And every one of them was told the answer while it learned.

Today, We Will Answer Without Key

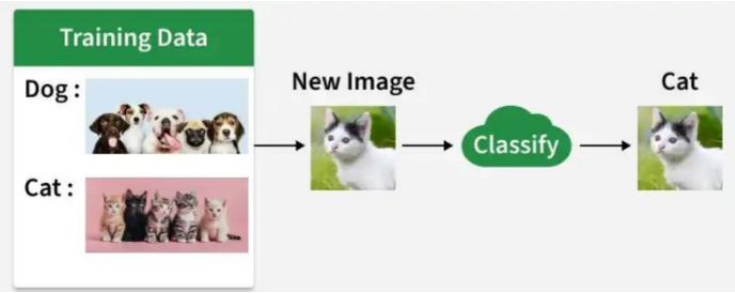
- Today we take the answer key away.
- Nobody tells the algorithm what the groups are. It has to find them.

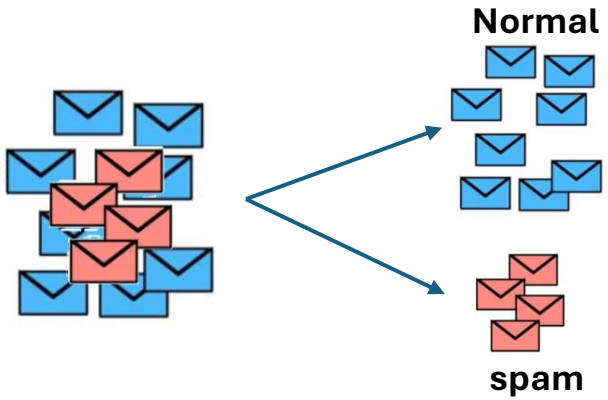
What is Clustering?

Supervised to Unsupervised

- **Everything until now was SUPERVISED.**
  You had a labelled column. The model learned the pattern that produced it.
  You could check the answer, because you had the answer.

- **Clustering is UNSUPERVISED.**
  There is no label. There is no target column. Nothing to predict.
  The algorithm groups the rows by similarity, and then you decide what the groups mean.





What is Clustering?

Classification vs Clustering

	Classification	**Clustering**
Is there a label?	Yes. You are given the answers	No. There is no target column
What it learns	The pattern that produces the label	Which rows look like each other
Number of groups	Fixed. You know the classes	Unknown. Sometimes you must guess it
How you score it	Accuracy, precision, recall, AUC	No accuracy exists
Who names the groups	Somebody already did	You do, after you look at them
Can it be wrong?	Yes, and you can prove it	It can be useless, but not wrong

What is Clustering?

Two Questions, Same Houses

- **CLASSIFICATION asked:**
  "Is this house above the median price?"
  We already knew the answer for 2,930 houses. The model learned to repeat it.
  Success is measured: 90 percent accuracy.

- **CLUSTERING asks:**
  "Do these 2,930 houses fall into natural groups? What are they?"
  Nobody knows the answer. There is no accuracy to compute.
  Success is measured by whether the groups mean something to a realtor.

One question has a right answer. The other has a useful answer.

What is Clustering?

Three Problems You Did Not Have Before

- **There is no accuracy to compute.**
  No labels means no confusion matrix, no precision, no recall.

- **Every algorithm always returns groups.**
  Feed it pure random noise and it will still hand you clusters. Confidently.

- **You have to name them yourself.**
  The output is “cluster 0, 1, 2.” Turning that into “starter homes” is your job.

So the real question today is not how to run it.
It is how you know the groups you found are real.

k-means Clustering

Definition

k-means partitions the objects into k clusters, each represented by its centroid, and assigns every object to the cluster whose centroid it is closest to.

Chapter 7
Introduction to Data Mining, 2nd Edition

Where to use ?

- **Customer segmentation:** Grouping shoppers by what and how much they buy.
- **Image compression:** Reducing millions of colors to 16.
- **Document grouping:** Sorting articles into topics nobody labelled.

Pros?

- **Fast.** Scales to very large datasets with no trouble.
- **Simple to explain.** Each cluster is described by one centroid.

Cons?

- You must choose k before you start, and you rarely know it.
- It assumes clusters are round and similar in size. Real groups often are not.
- Sensitive to the starting positions. Different runs, different answers.

k-means Clustering

The Working of k-mean Clustering

Find groups of people based on their age and spending amount

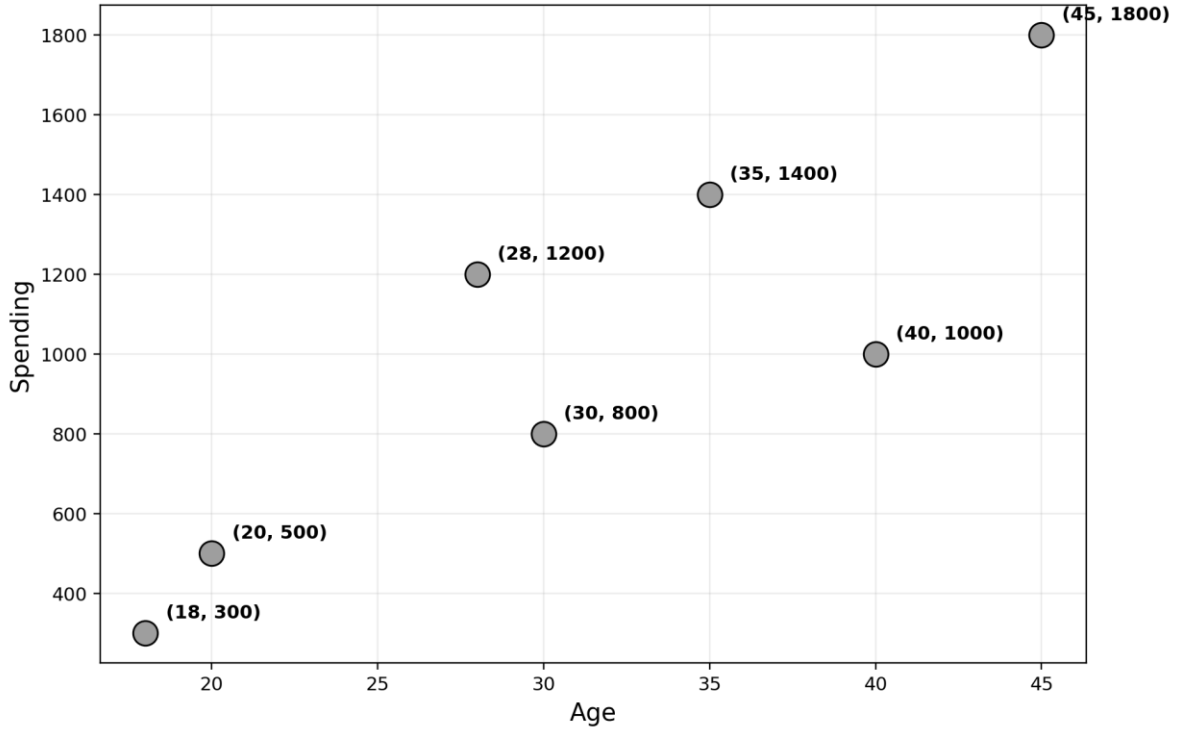
Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800

Step 1: Choose the number of clusters (K)
Step 2: Initialize K random centroids
Step 3: Assign each data point to the nearest centroid
Step 4: Update the centroid positions
Step 5: Repeat until convergence

k-means Clustering

The Working of k-mean Clustering

Age	Spending
20	500
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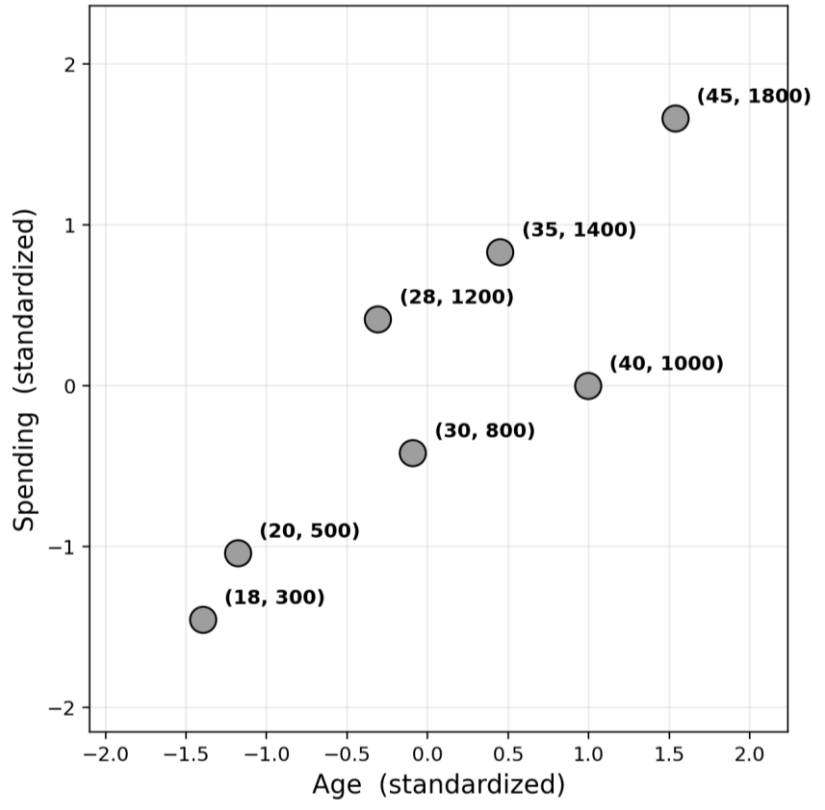


k-means Clustering

The Working of k-mean Clustering

Age	Spending
20	500
40	1000
30	800
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Present the data in standard scaling



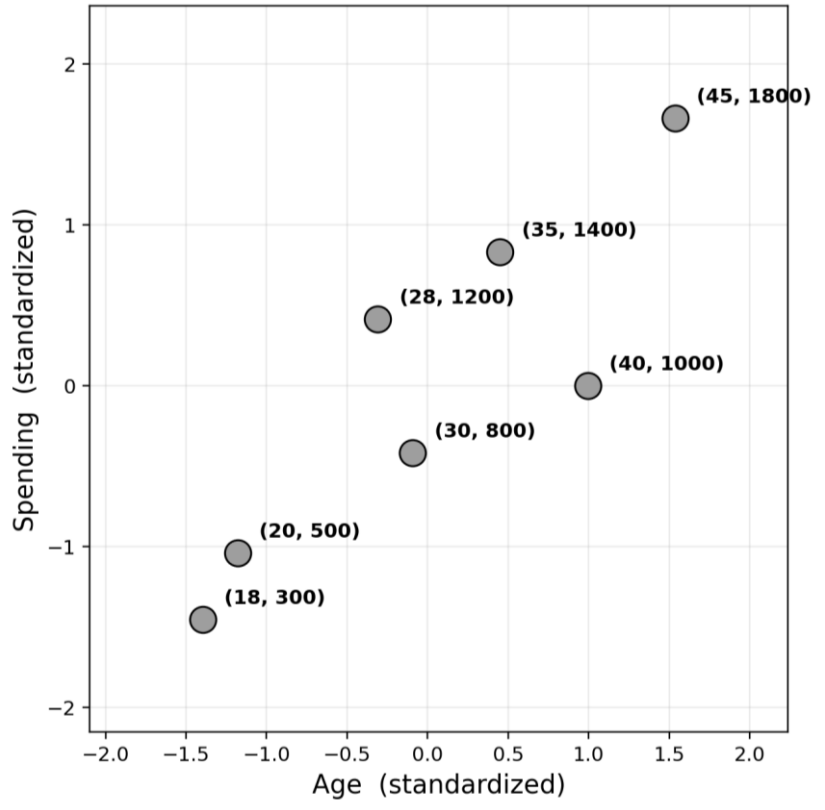
k-means Clustering

The Working of k-mean Clustering

Step 1: Choose the number of clusters (K)

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800

K = 2

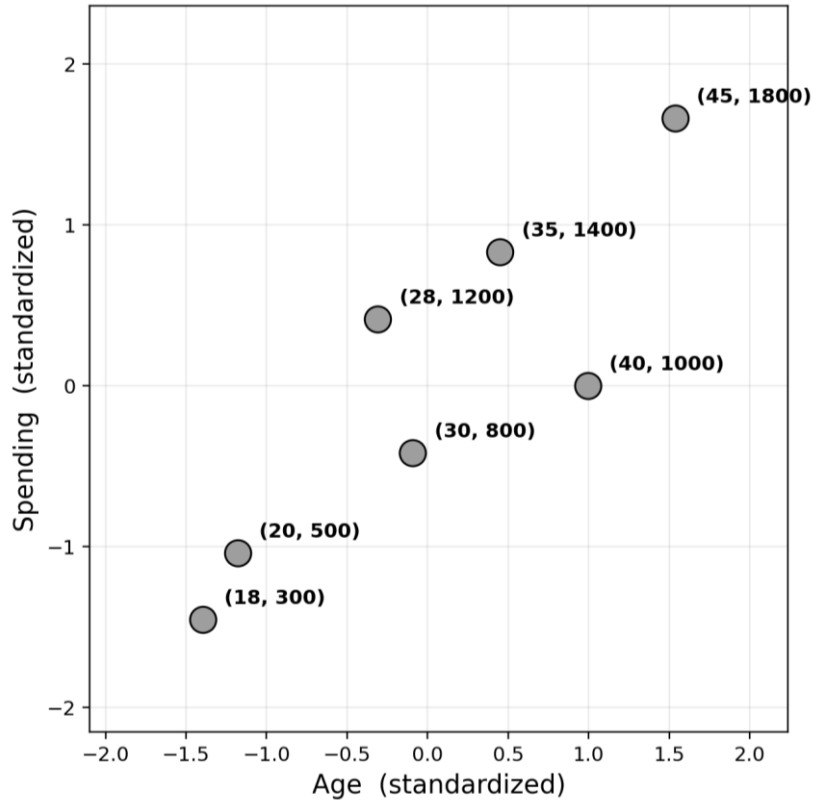


k-means Clustering

The Working of k-mean Clustering

Step 2: Initialize K random centroids

Age	Spending
20	500
40	1000
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18	300
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45	1800

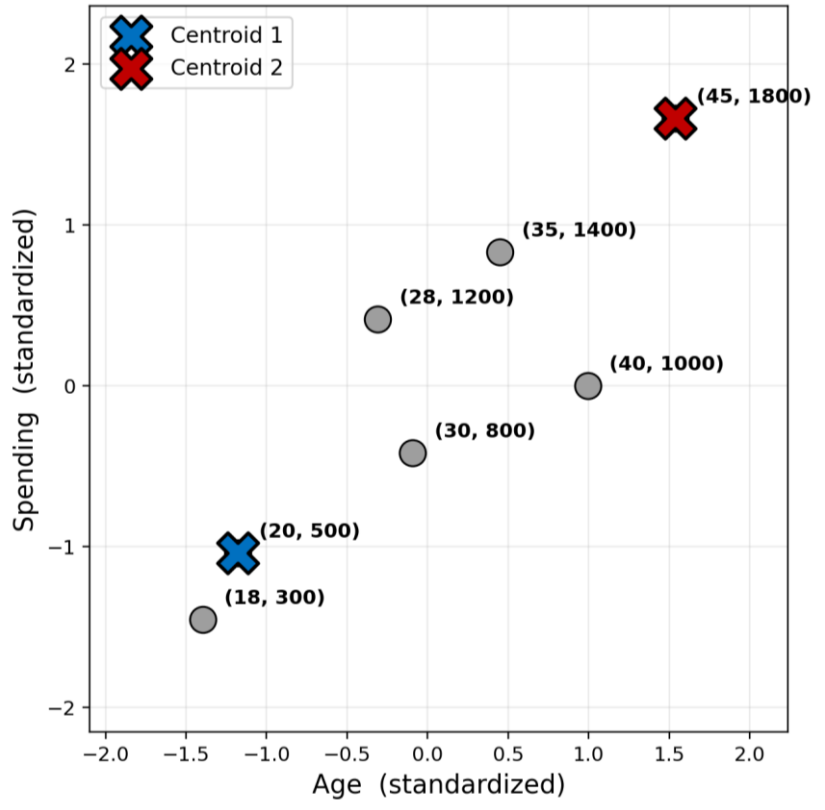


k-means Clustering

The Working of k-mean Clustering

Step 2: Initialize K random centroids

Age	Spending
20	500
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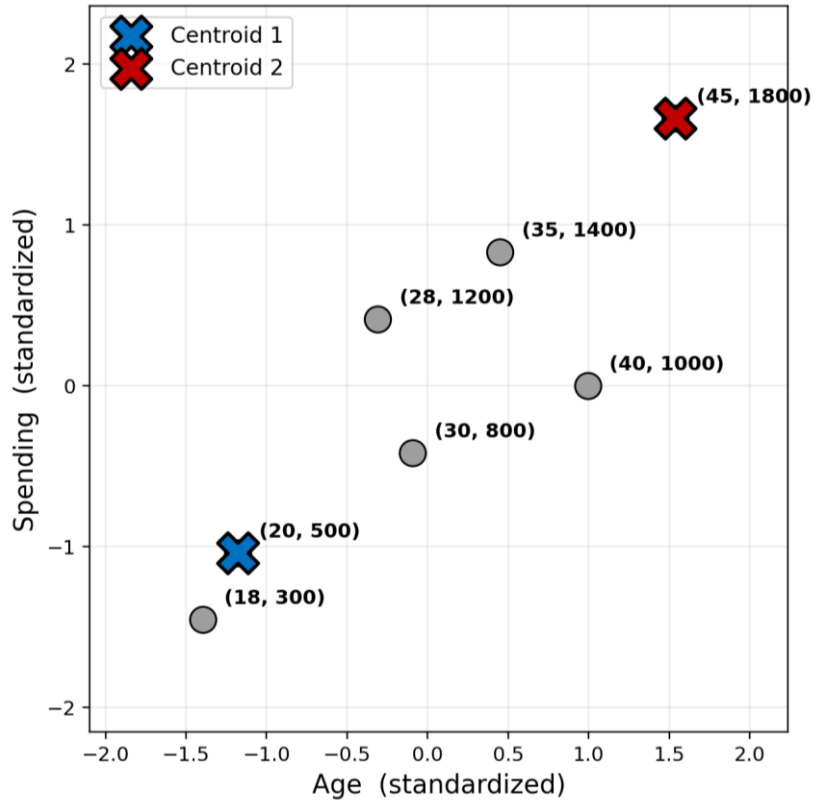


k-means Clustering

The Working of k-mean Clustering

Step 3: Assign each data point to the nearest centroid

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800



k-means Clustering

The Working of k-mean Clustering

Point	Raw	Standardized
P	(30, 800)	(−0.093, −0.416)
Centroid 1	(20, 500)	(−1.180, −1.039)
Centroid 2	(45, 1800)	(+1.537, +1.663)

Step 3: Assign each data point to the nearest centroid

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)}$$

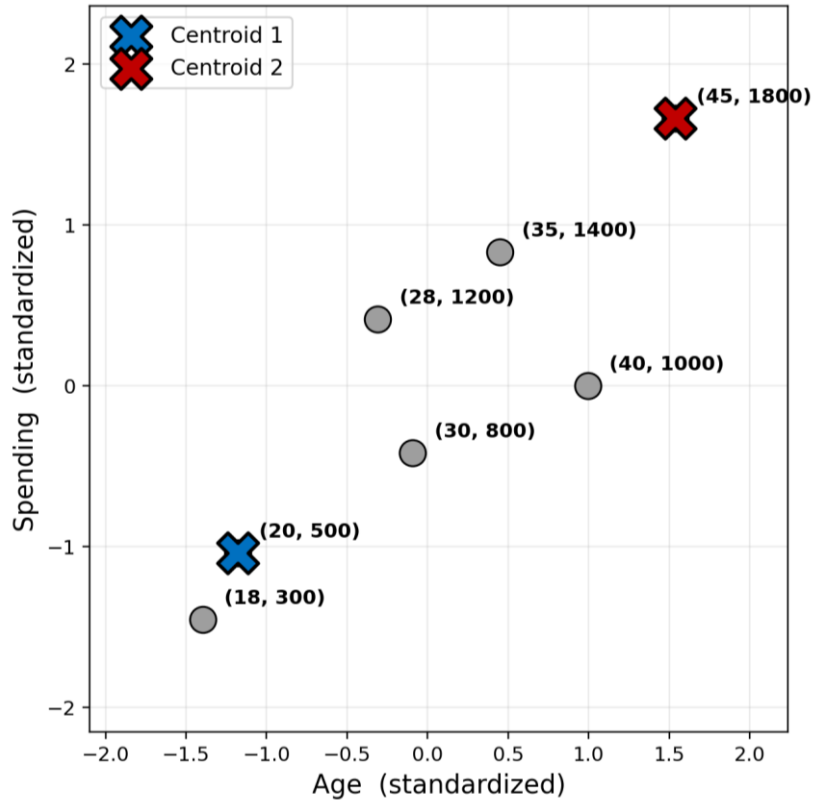
$$d=\sqrt{(-0.093-1.537)^2+(-0.416-1.663)^2}$$

$$d=\sqrt{(-1.630)^2+(-2.079)^2}$$

$$d=\sqrt{2.657+4.321}$$

$$d=\sqrt{6.978}$$

$$d=2.64$$



k-means Clustering

The Working of k-mean Clustering

Point	Raw	Standardized
P	(30, 800)	(−0.093, −0.416)
Centroid 1	(20, 500)	(−1.180, −1.039)
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$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)}$$

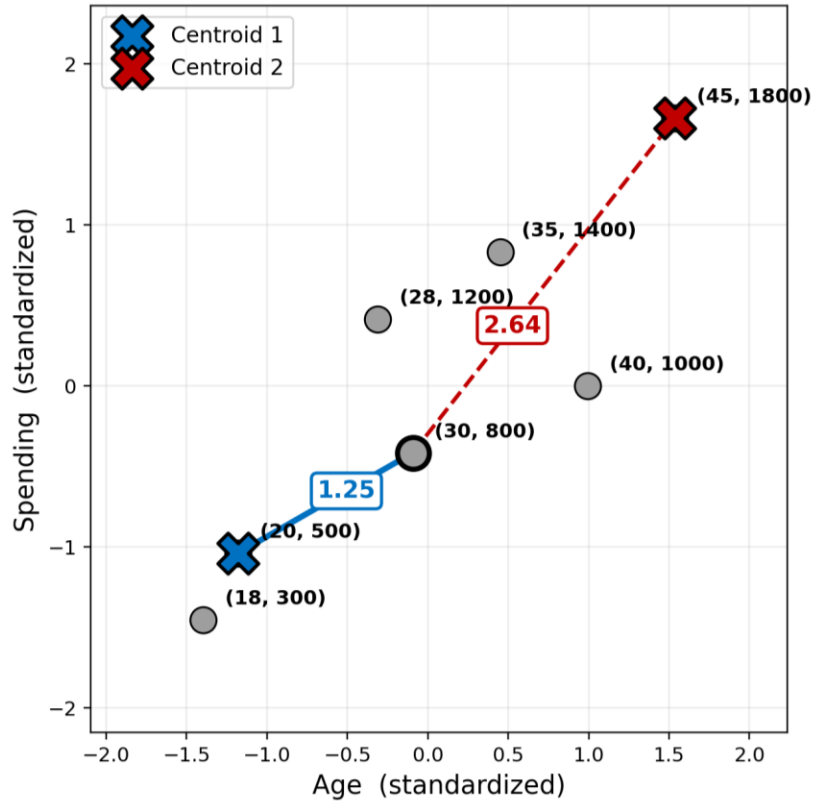
$$d=\sqrt{(-0.093-1.537)^2+(-0.416-1.663)^2}$$

$$d=\sqrt{(-1.630)^2+(-2.079)^2}$$

$$d=\sqrt{2.657+4.321}$$

$$d=\sqrt{6.978}$$

$$d=2.64$$

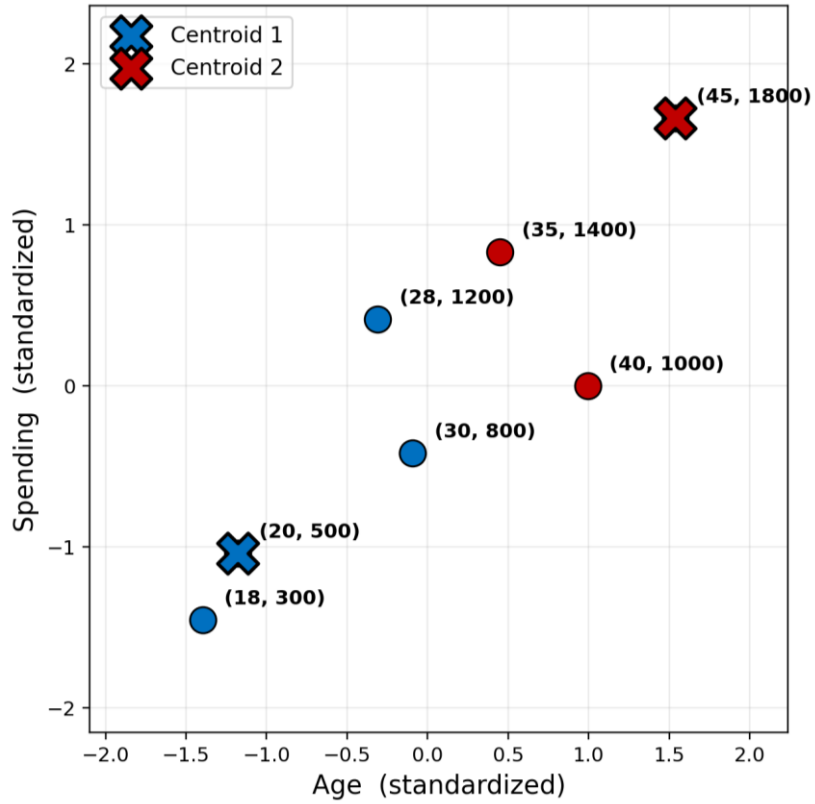


k-means Clustering

The Working of k-mean Clustering

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800

One iteration done !



k-means Clustering

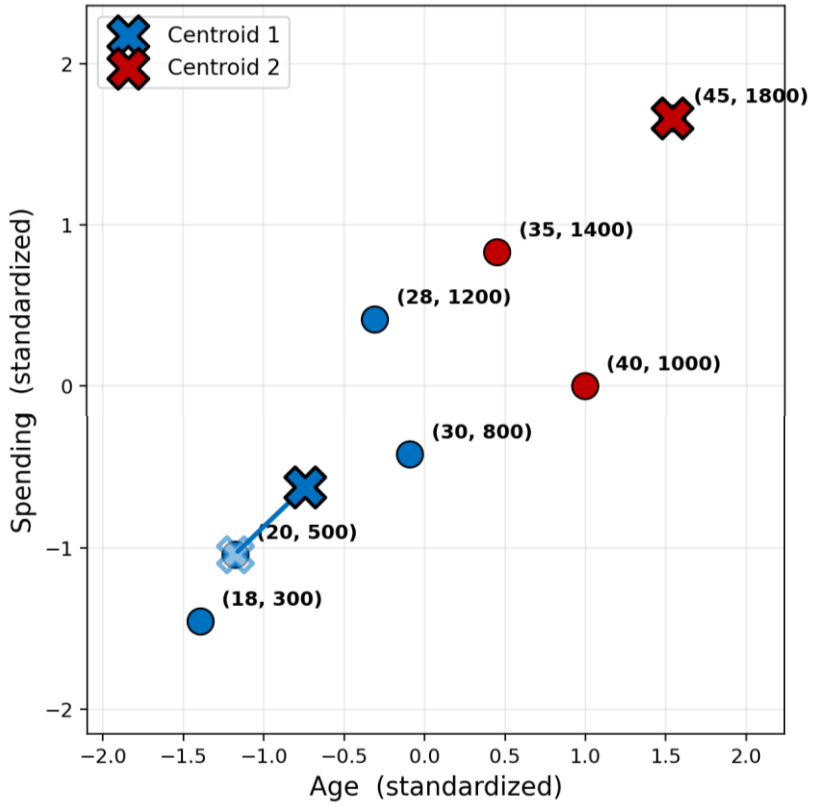
The Working of k-mean Clustering

Step 4: Update the centroid positions

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800

Centroid 1 (blue)

Point	Standardized
(20, 500)	(−1.180, −1.039)
(30, 800)	(−0.093, −0.416)
(18, 300)	(−1.397, −1.455)
(28, 1200)	(−0.311, +0.416)
Sum	(−2.980, −2.494)
÷ 4	**(−0.745, −0.624)**



k-means Clustering

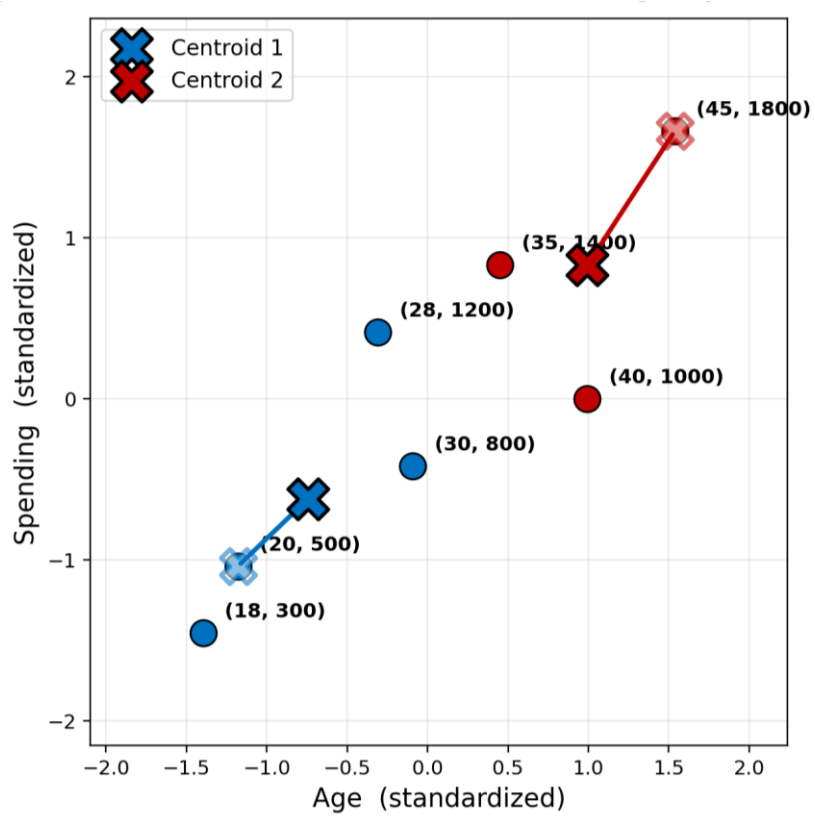
The Working of k-mean Clustering

Step 4: Update the centroid positions

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800

Centroid 2 (red)

Point	Standardized
(40, 1000)	(+0.994, 0.000)
(35, 1400)	(+0.450, +0.832)
(45, 1800)	(+1.537, +1.663)
Sum	(+2.980, +2.494)
÷ 3	**(+0.994, +0.832)**

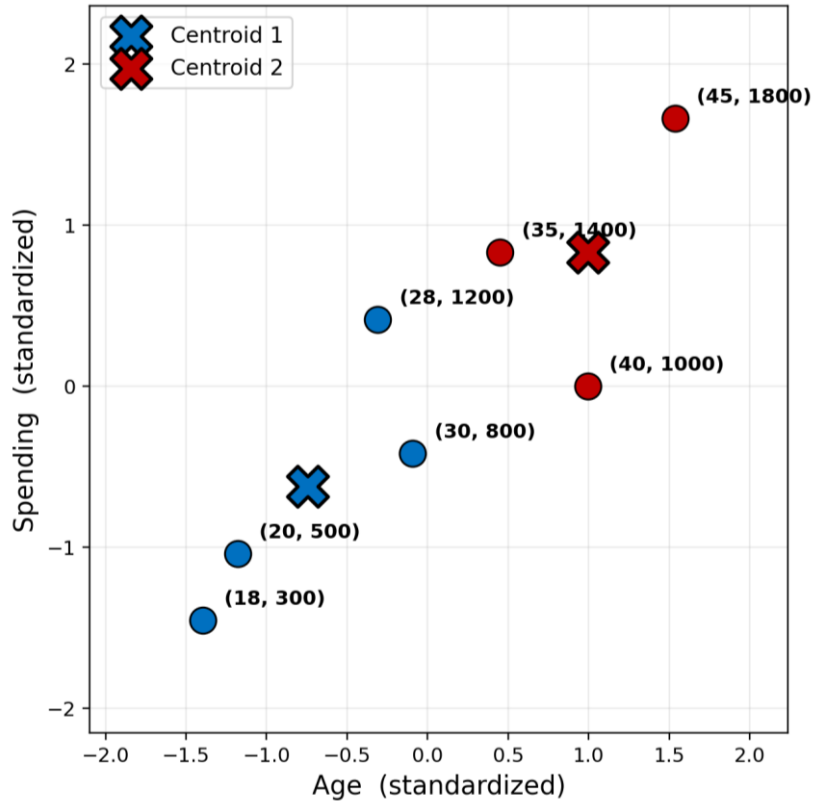


k-means Clustering

The Working of k-mean Clustering

Step 5: Repeat until convergence

Age	Spending
20	500
40	1000
30	800
18	300
28	1200
35	1400
45	1800



k-means on Ames

Our Question, and What the Code Will Do

- **The question**

  “The assessor has no price bands yet. Split these 2,930 houses into 5 groups.
  What are those groups, and would a realtor recognise them?”

- **What the code will do**

  Scale the 36 numeric columns first.
  Run k-means for k = 2 up to 10 and plot the elbow.
  Pick a k, then describe each cluster by its average house.
  Give every cluster a name a realtor would use.

- **k-means will give you exactly 5 groups. Whether 5 is right is your problem.**

Hierarchical Clustering

Definition

Hierarchical clustering produces a set of nested clusters organized as a tree, built either by repeatedly merging the closest pair, or by repeatedly splitting.

Chapter 7
Introduction to Data Mining, 2nd Edition

Where to use ?

- **Biology:** The evolutionary tree of species is a dendrogram.
- **Gene expression:** Grouping genes that switch on together.
- **Document taxonomies:** Topics inside topics inside topics.

Pros?

- You do not have to choose the number of clusters in advance.
- The dendrogram shows you every possible number at once. Cut it where you like.
- Deterministic. Same data, same answer, every run.

Cons?

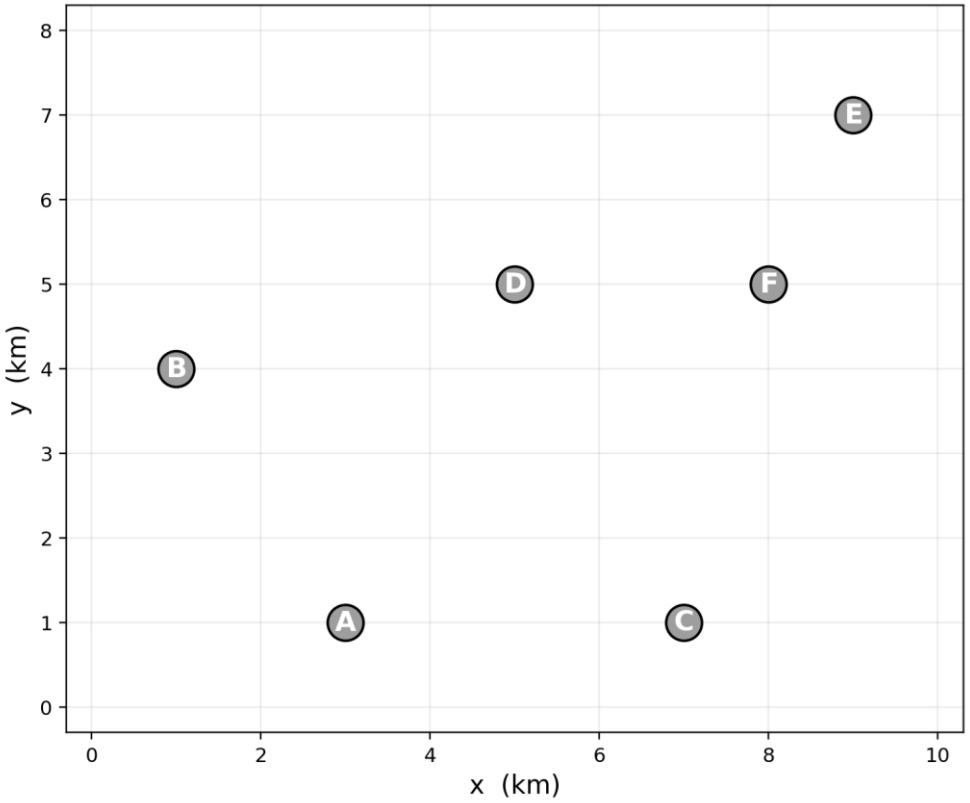
- Slow. Comparing every pair does not scale past a few thousand rows.
- Once two clusters merge, they never separate. An early mistake is permanent.
- The linkage choice changes the answer a lot, and there is no correct one.

Hierarchical Clustering

The Working of Hierarchical Clustering

Six coffee shops in a city.
A delivery company wants to group them into delivery zones.

Shop	x	y
A	3	1
B	1	4
C	7	1
D	5	5
E	9	7
F	8	5



Hierarchical Clustering

The Working of Hierarchical Clustering

Shop	x	y
A	3	1
B	1	4
C	7	1
D	5	5
E	9	7
F	8	5

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)}$$

$$d=\sqrt{(9-8)^2+(7-5)^2}$$

$$d=\sqrt{(1)^2+(2)^2}$$

$$d=\sqrt{1+4}$$

$$d=\sqrt{5}$$

$$d=2.236$$

$$d=2.24$$

Distance between any 2 shops

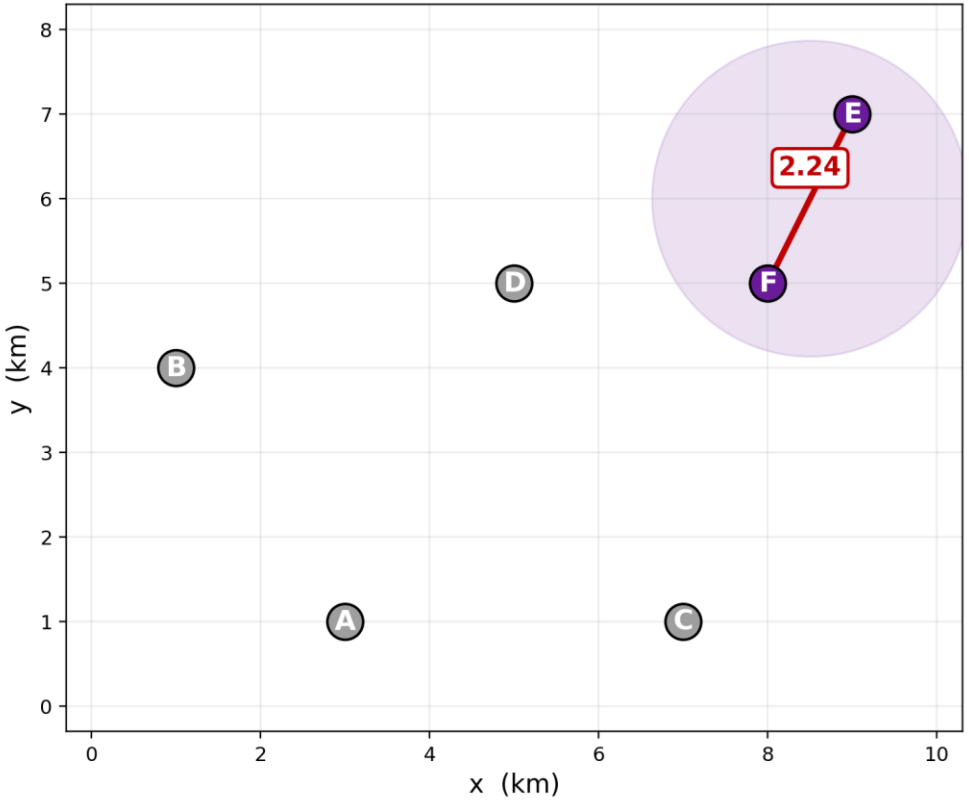
	A	B	C	D	E	F
A	–	3.61	4.00	4.47	8.49	6.40
B	3.61	–	6.71	4.12	8.54	7.07
C	4.00	6.71	–	4.47	6.32	4.12
D	4.47	4.12	4.47	–	4.47	3.00
E	8.49	8.54	6.32	4.47	–	2.24
F	6.40	7.07	4.12	3.00	2.24	–

Hierarchical Clustering

The Working of Hierarchical Clustering

Shop	x	y
A	3	1
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2.24



Hierarchical Clustering

The Working of Hierarchical Clustering

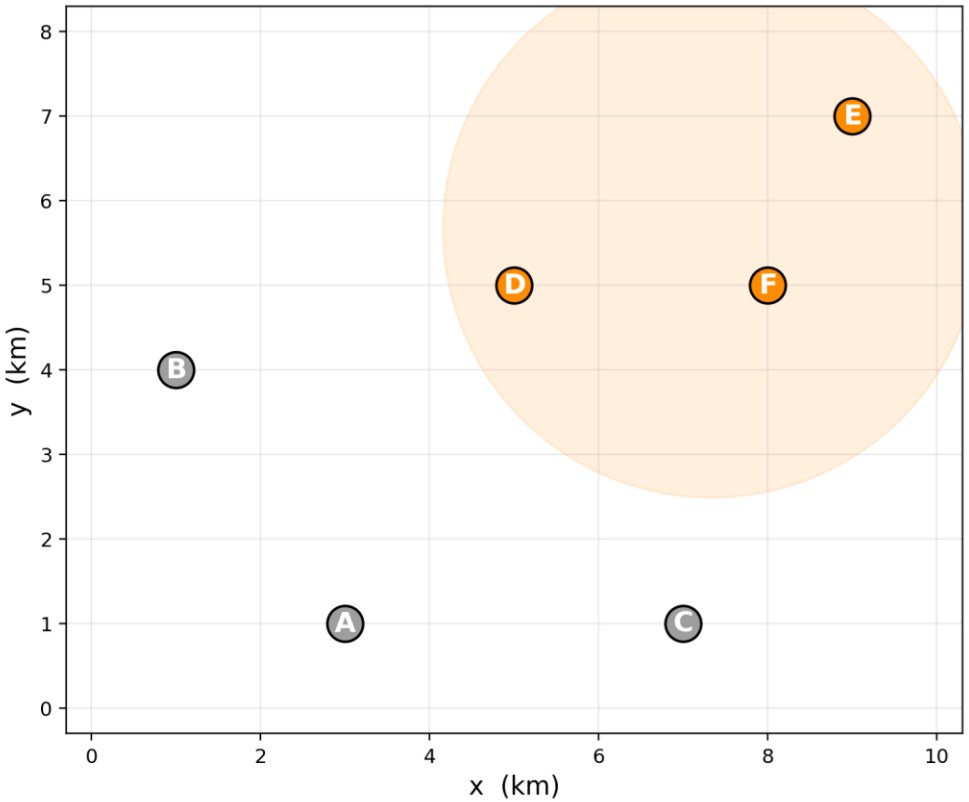
	How it measures group distance
Single linkage	Closest pair of members
Complete linkage	Furthest pair of members
Average linkage	Mean of all pairs

Shop	x	y
A	3	1
B	1	4
C	7	1
D	5	5
E	9	7
F	8	5

3

2.24

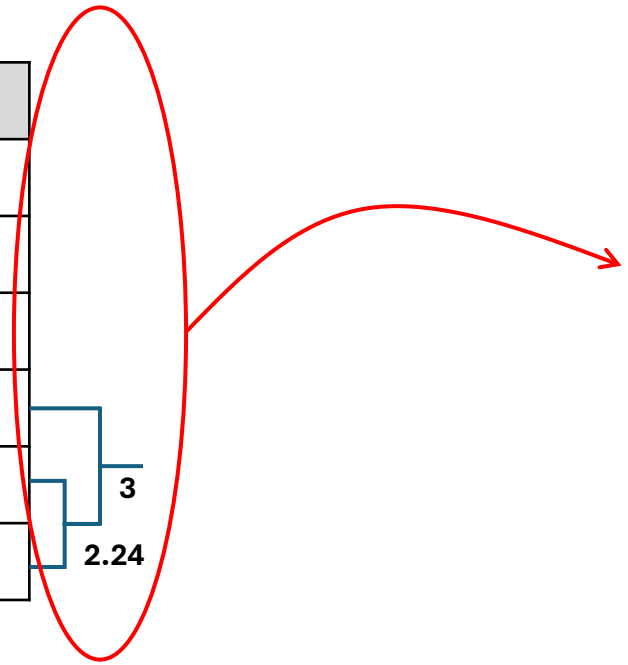
Single: min(4.47, 3.00) = **3.00**
Complete: max(4.47, 3.00) = **4.47**
Average: (4.47 + 3.00) / 2 = **3.74**

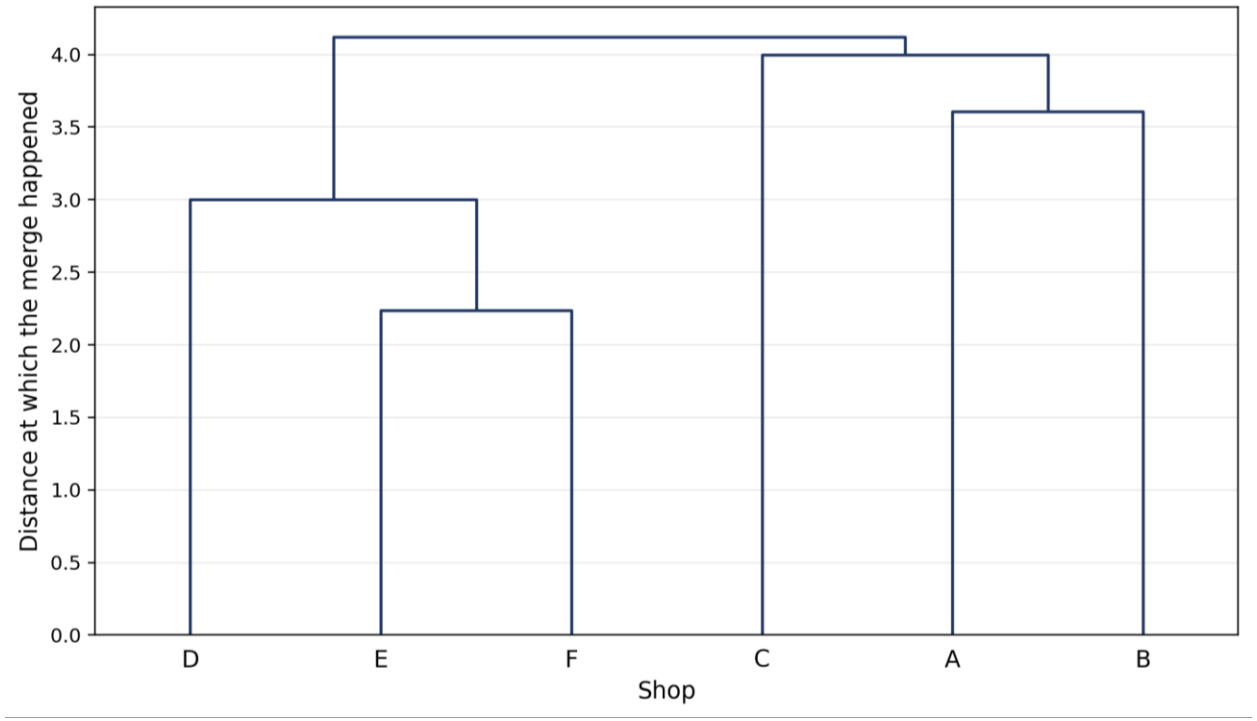


Hierarchical Clustering

The Working of Hierarchical Clustering

Shop	x	y
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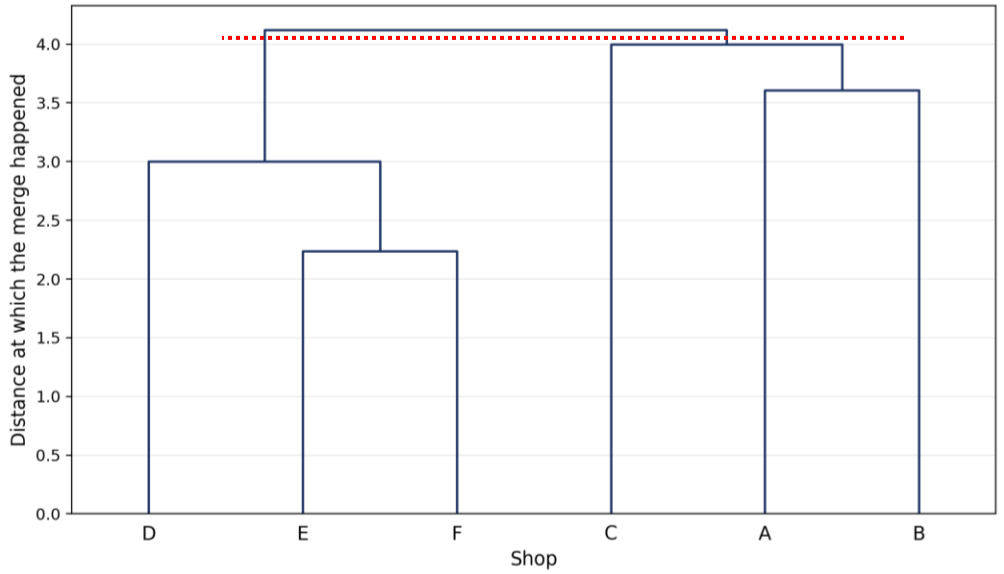


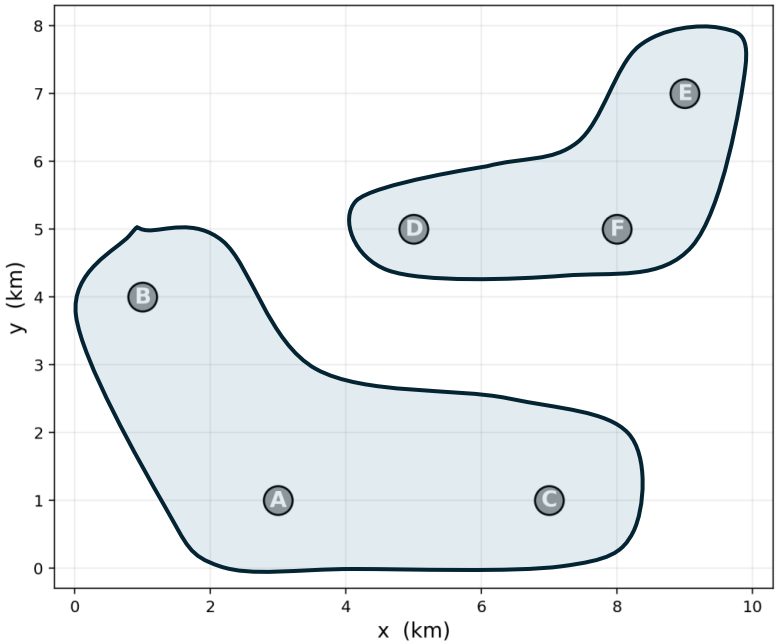


Hierarchical Clustering

The Working of Hierarchical Clustering

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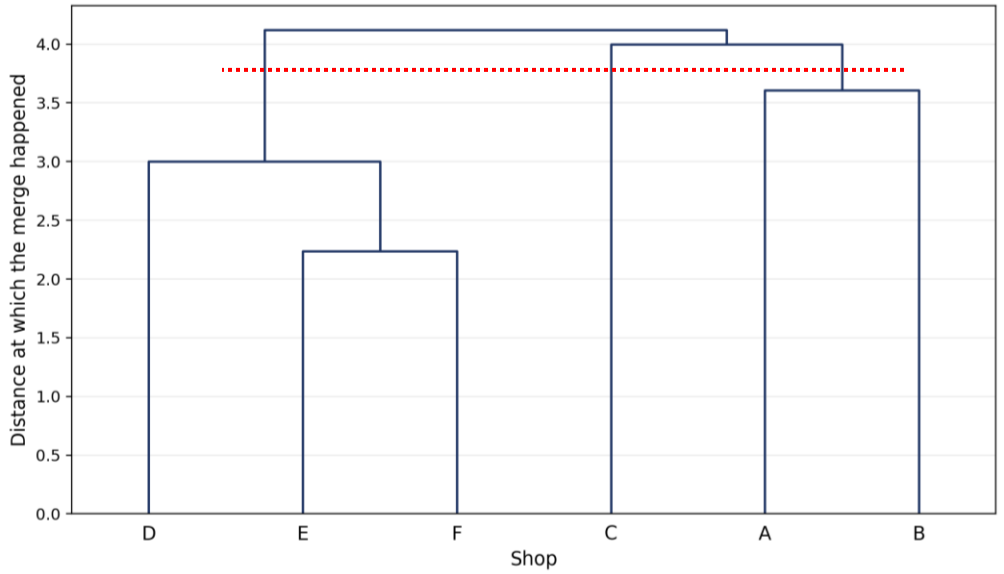


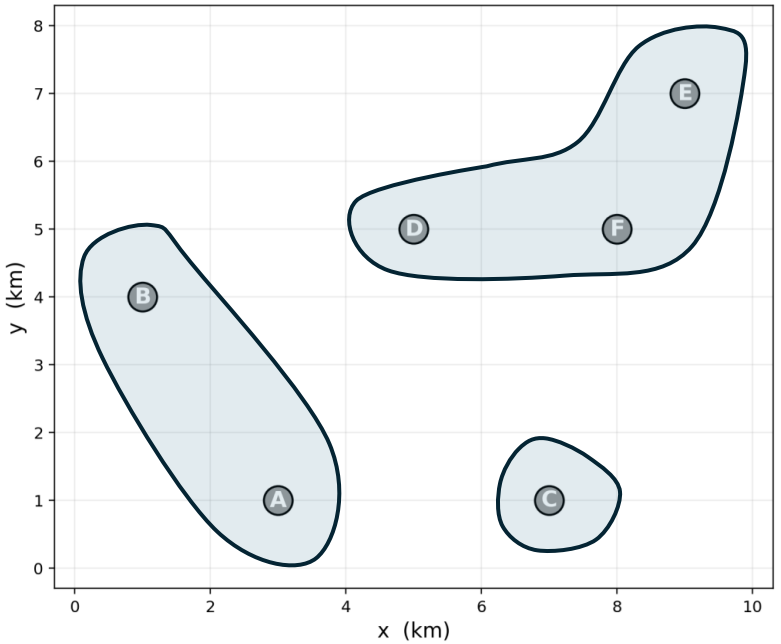


Hierarchical Clustering

The Working of Hierarchical Clustering

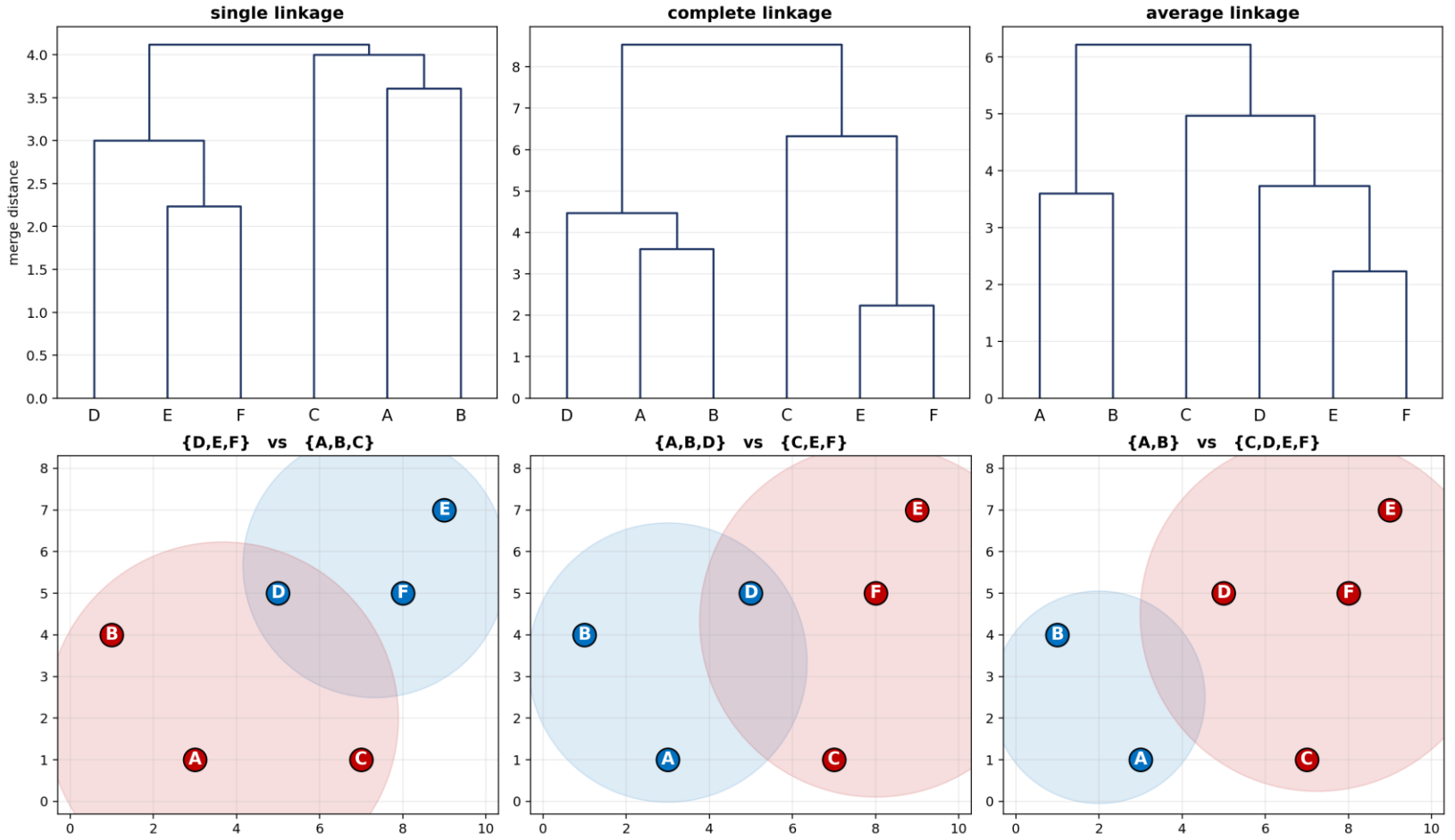
Shop	x	y
A	3	1
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D	5	5
E	9	7
F	8	5





Hierarchical Clustering

The Working of Hierarchical Clustering



Hierarchical on Ames

Our Question, and What the Code Will Do

- **The question**

  “Instead of choosing a number, build the whole tree of these houses.
  Then look at it and decide where it naturally wants to be cut.”

- **What the code will do**

  Take a sample, because 2,930 rows makes an unreadable dendrogram.
  Build the tree and draw it.
  Try three linkage methods and watch the answer change.
  Cut the tree at different heights and compare against k-means.

- **Same houses, same distances. A different answer because the rule changed.**